

Internet phone — changing the telephony paradigm?

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This paper provides an overview of the history, current status, and underlying technology of 'Internet telephony' — the transport of speech information over a packet-switched connectionless network. The economics of Internet telephony and a number of possible system architectures for interconnection with the switched telephone network are examined. The paper concludes with a look at the quality of service issues raised by Internet telephony.

1. Introduction

Internet telephony is the use of a packet-switched data network (most usually the global Internet, but it has certainly been implemented on other packet networks) for the transport of speech information between the participants in a real-time conversation. As such it is a major departure from the traditional use of such networks, which have been designed and optimised for the delivery of textual and other non-real-time data with guaranteed integrity, regardless of the time required to do so.

This paper is intended as an overview of Internet telephony technology as it is today. It does not consider the implications of changes to the Internet network itself [1], nor to the computers used to access the network (such as the use of network computers). It is a technical and economic discussion of currently deployed products and services and does not imply that BT has any proposals or plans for products or services in this field.

2. Background

The first documented Internet telephony experiments were conducted on the then ARPANET (the forerunner of the Internet) by researchers at MIT in the mid 1970s, resulting in the publication of an Internet protocol specification, RFC741 [2], for the 'Network Voice Protocol', in 1977. This specification documents most of the features of a modern Internet phone implementation and cites work as early as 1973. It is likely that these early experiments were related to the original role of the ARPANET/Internet as a prototype for a highly survivable military communications network.

Early audio experiments in audio transmission on packet networks were limited to academic environments, since audio data could not be compressed much below the 64 or 56 kbit/s of telephony PCM by the computers then available, and sound input and output devices had to be made because there were none to be bought.

Packet audio was hardly heard of again (outside the field of secure radio) until the computing power to compress speech sufficiently to be transmitted over the limited bit rate of a dial-up connection became affordable. By 1993 the computing power to compress to 14.4 kbit/s or below (a fast 486 PC) and the modems to connect at that rate became affordable, and the first commercial Internet phone applications appeared. The early I-Phones evolved either as spin-offs from the videoconference industry or from companies like Vocaltec which, based in both Israel and the USA, sought to use in-house expertise to reduce their international call charges and then realised that the resulting software was a saleable product.

More recently, all of the major desktop software and hardware vendors (Netscape, IBM, Intel and Microsoft) have made available 'free' I-Phones, bundled with their operating systems or World Wide Web browsers. Microsoft have now gone further and made available a 'System Developer's Kit' containing the components of their NetMeeting™ Internet Phone and Whiteboard for anyone to use as the basis of their own Windows95™ conferencing application.

These programs were only capable of connecting users who owned a PC with a sound card and a fast (14.4 kbit/s or better) Internet connection. As such, their user community and commercial significance remained limited until the summer of 1996, when two US-based dial-back operators launched services allowing registered users to connect over the Internet to their gateways in the USA from anywhere in the world, the call being completed over the PSTN from their gateway. This is essentially an alternative to the longer-established 'dial-back' method of bypassing normal international call rates.

At the time of writing in late 1996, the first signs of standardisation (which may eventually lead to interworking between the various products) are beginning to be seen. The industry appears to be converging on H.323 [3], a suite of protocols developed by the ITU-T (formerly CCITT) for LAN-based videophones, but configured for audio and data components only. Overviews of the current H.323 and audio compression standards can be found in Thom [4], and Cox and Kroon [5].

3. Technology

3.1 The Internet network

The Internet was designed for datagram traffic. Datagrams are fragments of a message or file, each of which has attached a source address, destination address, and sequence number. Each datagram is independently routed across the network and the message or file must be reassembled at the receiving end from the received fragments. This is rather like the 19th century Morse telegraph network — especially if one imagines a long message being sent as a series of individual telegrams. It was entirely normal for a telegram to be written down and retransmitted at several points between the sending and receiving telegraph offices. The Internet works by forwarding datagrams from computer to computer in a similar way.

The Internet was designed to deliver messages (datagrams) reliably regardless of the delay. The basic Internet protocol (IP) provides for the routing of these messages, typically up to about 1000 bytes in size, between any two network nodes, with checking for corruption and loss. A higher-layer protocol, the transmission control protocol (TCP) provides for the re-transmission of lost data. Retransmissions will be requested repeatedly until all of the required data has been transferred. Since the retries take some time (due to the round trip delay of the acknowledgement as well as simple retransmission) TCP can take much longer than 'size of message $\times$ data rate' to complete transmission. The probability of a retransmission being required is determined by network congestion (mostly) and the acknowledgement delay by the number of network nodes between the sending and receiving ends. As such TCP/IP is highly unsatisfactory for transmission of fixed-rate data.

An alternative higher-layer protocol, the user datagram protocol (UDP) can be carried by an IP network. UDP was historically used for short unacknowledged messages that could simply be repeated by the sender if not acted upon in what it considered to be a reasonable time. If the IP network was unable to deliver UDP datagrams they were simply discarded. UDP therefore offered some hope that the constant-rate sender could avoid a backlog of data in a congested network situation. The problem was that the

receiver could expect to receive packets with a variable transmission delay which can be anywhere from 100 to 1000 milliseconds. It could also be expected that some packets would be lost with the percentage loss being anything up to 30%.

The TCP protocol has an intrinsic rate-limiting mechanism — the more forwarding hops, the longer an acknowledgement takes. UDP has no such limiting mechanism, and therefore competes more effectively than TCP for the available bandwidth in a congested IP network.

Unlike the Internet, telephone communication has always assumed a direct, reserved physical connection (or at least a time-slot in a PCM system) between the sender and receiver for the duration of a call. For most of their history, telephone systems have had a fairly straightforward numbering system specified by CCITT Recommendation E.164/I.331 [6] which directly related to their physical structure. These differences result in several significant assumptions which hold true for circuit-switched telephony being inapplicable to the Internet, and vice versa.

- A PSTN circuit offers essentially guaranteed quality connections for the duration of a call once connected. If the call cannot be connected, busy tone is heard. An I-Phone will always connect if a path exists, but if the reliable throughput is less than that required for 100% packet delivery, quality will suffer. Demand is to some extent self regulating — as more users attempt calls over a given route, packet loss increases and quality suffers to the point that new users are deterred.
- One user or application cannot affect the quality of service seen by another PSTN user once a call has been connected. This is not true for the Internet — the available bandwidth is shared by many services (WWW, E-Mail, I-Phone, file transfers, etc) and an increase in use by one application can only be at the expense of another. The existence of a path with sufficient capacity at the start of an I-Phone call is no guarantee that it will continue to exist throughout the call.
- If Internet telephony technology is used to multiplex audio and data over a single circuit-switched connection (as in dial-up access to the Internet) then the quality of service for that access link can be as good as the circuit, providing that the sharing of bandwidth between audio and data can be effectively managed.

3.2 Implementation

An Internet phone (see Fig 1) requires several elements to work. These are:

- an audio input/output device (a sound card) and transducers (speaker and microphone),

- a LAN or dial-up connection to the Internet,
- an audio codec (software or hardware) capable of compressing speech from the sample rate of the sound card (typically 8K 16-bit samples per second = 128 kbit/s) to a lower rate that will fit the modem capacity, (see Cox and Kroon [5] for a detailed description),
- a processor of sufficient power to run the audio codec in real time (with some capacity to spare for Windows™ and other applications),
- a protocol stack for the delivery of real-time data over a packet network,
- a signalling protocol to permit the establishment, maintenance, and clearing of calls.

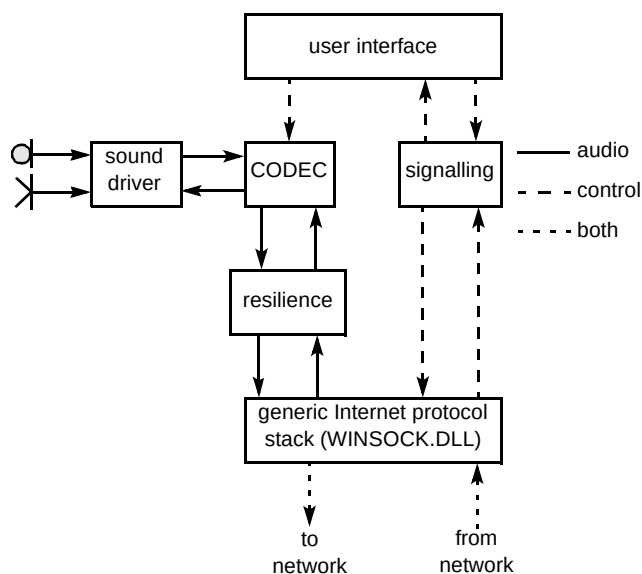


Fig 1 Internet phone software architecture.

The following sections will deal with these in more detail.

Audio input/output devices

An increasing number of personal computers have been equipped with sound cards over the last few years, to the point where they are being found in the office as well as in home machines. The majority are based on the original SoundBlaster™ developed by Creative Labs, and are so called 'half duplex' cards, which can record (send) or play (receive) but can not do both at the same time. More recently full-duplex designs (such as the Gravis Ultrasound™ have become available. These cards are able to record and play simultaneously, and are able to pass collected samples to the host PC more frequently, which makes them much more useful for telephony applications. Some cards which were originally half-duplex (including at least some of the Creative Labs SoundBlaster family now

have improved driver software which permits full-duplex operation.

Internet access

It is not entirely certain how many genuinely Internet-connected computers there are in the world. The claimed totals in excess of 40 million users can be broadly split into two groups — those who dial up a commercial service provider using a modem over ordinary telephone lines, and those who access the Internet over a local LAN with a shared upstream connection over some kind of leased line (small companies may have a LAN and a shared dial-up connection).

Currently, modems capable of rates up to 14.4 kbit/s, and telephone lines good enough to support connections at that rate, are commonplace, and the current 'leading edge' is 33.6 kbit/s, with 28.8 kbit/s modems the majority of new sales this year (1997). How many of these connect at full speed will depend on access line quality and is not accurately known, but it is probably safe to assume that an I-Phone has 14.4 kbit/s to use, minus the overheads of the Internet protocol (see below) and the capacity used by other Internet applications such as a WWW browser or e-mail being used at the same time. In practice, most I-Phones in the field have used coded audio bit rates around 8 kbit/s.

The software to implement the Internet protocols takes various forms depending on the computer and operating system in use. Unix™ machines have the Internet protocol built into their operating system. Windows applications use a dynamic linked library called WINSOCK.DLL which contains the necessary extensions to the Windows operating system (see Fig 2). Although individual WinSock implementations may be written for different network devices (such as a modem, Ethernet LAN card, or ISDN), all of them offer the same interface to Internet application programs, be they e-mail clients, browsers, or Internet phones.

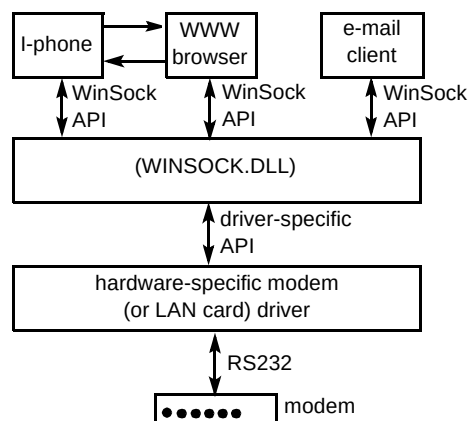


Fig 2 WinSock architecture.

Speech compression

Traditional telephone networks use pulse code modulation (PCM) at 8K samples per second. 12-bit samples are compressed and expanded ('companded') by a nonlinear look-up table into 8-bit words giving a transmitted rate of 8 kbit/s. The compression typically used by an Internet phone today is of the order of 16 to 1 (128 kbit/s to 8 kbit/s). Such compression is beyond PCM, ADPCM (32 kbit/s, used in CT-2 cordless phones), or sub-band coding (down to 16 kbit/s for speech bandwidths, normally used for music at higher bit rates). It is therefore necessary to use a 'vocoder' or a codec based on some kind of speech-specific model. By building more knowledge about the characteristics of human speech and hearing into the codec it is possible to get compression as good as 50:1 (US proposed federal standard mixed excitation linear predictive (MELP) codec [7]), at the price of the codec being totally unsuitable for any source material other than speech.

Most vocoders implement some form of linear predictive coder (LPC) algorithm. Such codecs analyse incoming speech, and transmit the parameters required by a voice synthesiser at the receiving end to speak the same syllables. The algorithms vary in the number of synthetic voice parameters sent, the choice of parameters, and the way in which they are encoded for transmission.

All of these codecs were designed for constant bit rate applications (such as digital mobile phones or videophones) and require additional software to protect them from the variable data rate offered by a packet network, as well as actual packet loss.

Examples of these are the codec used in GSM mobile phones [8] that used to compress messages stored in BT's Call-Minder service [9], and the US Federal Standard 1016 CELP [10] codec used for secure radio systems. Cox and Kroon [5] give a good overview of the currently popular coding algorithms. The majority of Internet phones in the field have used the GSM codec but the encoded data is stripped of its error-correction bits, which leaves a bit rate around 8 kbit/s to be transmitted in each direction for a full-duplex call.

A further significant reduction in average bit rate required comes from voice activity detection (this is often enforced by the use of half-duplex sound cards, but is generally desirable). Voice activity detection suppresses transmitted data either entirely or to occasional 'no data' messages when the user of a PC is not speaking. Since it is more likely that one person, the other person, or no-one is speaking than that both people will be speaking, average bandwidth is reduced by at least 50%.

Compression is of course entirely unnecessary for communication between users connected by a network of

sufficient bandwidth such as the same LAN segment. Few practical implementations have been uncompressed, however, because they are either limited to making very local calls or require a gateway with compression to make calls to other sites.

Processor

The primary demand placed on a computer by Internet phone software is due to audio compression. Audio compression software has traditionally been developed for digital signal processors (DSPs), which are optimised for the fast execution of relatively small programs on relatively small data sets, often with some degree of parallel processing. The programs are often highly repetitive and are mathematically demanding.

This is very different from the normal work of a desktop PC, which involves the movement of large quantities of data, which is seldom all in memory at once, and the programs themselves are generally decision-intensive rather than mathematically challenging (if they are not waiting for user input).

General-purpose microprocessors have not, therefore, evolved instruction sets well suited to speech compression, and have therefore only recently been able to run audio compression software in real time. The minimum specification for an 8 kbit/s CELP derived codec is approximately a 33 MHz 80486. The encoder for CELP-derived codecs typically requires around three times as much CPU time as the decoder. This is about as favourable a ratio as has been achieved. Rival algorithms such as fractal codecs (which require more encoder processing power but less to decode) have therefore found relatively few conversational applications.

Intel has announced an extension to the Pentium[®] processor instruction set for DSP-like programs. These MMX[™] instructions may lead to improved audio codec performance for a given Pentium clock frequency.

Resilience and delay

Internet phones generally use UDP transport, backed up by time stamps on each datagram (sometimes called a 'packet') and a large receive buffer. The combination of the time stamps and a receive buffer allows the receiving end to assemble the received audio packets in the correct order with the correct relative timing before they are played back. In the event that packets have not turned up by the time the receive buffer runs out they are replaced locally by the receiver. The perceived quality of one I-Phone relative to another is largely down to the implementation of this process. There is an IETF standard, the real time protocol (RTP) [11], which covers the time stamping and content type identification of UDP audio packets. The use of RTP is

generally limited to H.323-compatible products. Other products have used proprietary schemes which have various advantages and disadvantages relative to RTP, which was originally designed for multicast ‘broadcasting’ at higher bit-rates with combined audio, video, and data payloads. The use of RTP and related protocols for multimedia services is covered in Rudkin [12].

The receive buffer is typically of the order of 0.2 to 1 second in length. In a few products it is adjustable. The receive buffer needs to be longer than the worst expected difference in propagation times between sender and receiver. The substantial transmission delay observed with Internet phones is largely due to the presence of this buffer.

The action taken when a packet is lost may be any of:

- repeat the last packet,
- mute the output,
- insert a predicted or synthesised ‘fill-in’ packet.

The best action to take depends on the duration of the lost packet(s). Work done at UCL by Hardman, Sasse, Handley and Watson [13] suggests that packet repeats are the optimum solution for gaps of 20 ms or less for which they sound like drawn-out syllables. Over longer periods they sound more like a stammer and are subjectively less acceptable than muting.

An alternative to receive-end detection and replacement of missing packets is to provide some redundancy in the transmitted data stream. This may take the form of distributing the data from each sample period over a number of UDP packets, classical forward error correction (FEC), or a back-up audio stream transmitted at a lower rate. In the latter case, data relating to the same time period from the main and back-up streams is never sent together in the same packet (and therefore never lost together). This is the basis of work at University College, London on reliable audio transport [14] and it is understood that at least one commercial Internet phone product distributes bits from each audio sampling period over a number of UDP packets for this reason.

Call-control protocols

An Internet phone requires a signalling system that offers comparable behaviour to a circuit-switched network while using the datagram-oriented Internet.

In practice, an Internet phone signalling system must perform three tasks:

- find the IP address of the destination PC — this is not necessarily easy, since most dial-up IP users have an IP address allocated from a pool when they connect, which is reallocated to another customer afterwards,
- having found the IP address, establish communication with the called party’s PC and find out if it is capable of communicating using the caller’s Internet phone software,
- if communication is possible, the signalling system must negotiate the IP, audio compression, and resilience settings to be used for the call, and then perform the normal alerting-answered-connected-hangup-disconnected sequence of a PSTN phone call.

The requirements for the signalling system are further complicated if there is a requirement to communicate with a gateway between the Internet and the PSTN. A specific problem is that whereas the PSTN has a concept of a call which occupies a circuit and is charged for per minute or per second, the Internet treats each packet individually (and assumes that the sender has paid a subscription in advance, so has no metering). Internet applications do not therefore have an explicit ‘disconnect’ command in every case, and frequently use inactivity timers as well or instead. Many proposed Internet call-control protocols appear to have a weakness in this respect.

The signalling system is the main point of difference between current I-Phone applications. Although the industry appears to be converging on a derivative of the ITU-T standard H.323, this will not be completely solved, since the signalling component of H.323 (H.245) really only specifies the encoding and formatting of the data to be sent after a link has been established. More significantly, H.323 is based on a call-control model inherited from dial-up (ISDN) videoconference terminals, and does not fully exploit the multicast capabilities of a packet network. It is therefore likely that it will have to evolve or be replaced in order to address these limitations.

Current solutions for address discovery and call set-up communications include an Internet relay chat (IRC) server, manual entry of IP addresses, registration with a central directory (probably the most common solution), and use of the Internet domain name service.

Both reliable (TCP/IP) and unreliable (UDP/IP) protocols have been used for the signalling connection in Internet telephony. The reliable solution has the advantage that the signalling protocol can assume reliable lossless transmission — and can therefore be relatively simple), but does not scale up to large number of users as easily as an unacknowledged, unreliable solution based on UDP/IP, because it is necessary to maintain the protocol state for each separate signalling connection throughout a call. Most documented multicast signalling system proposals provide

reliability at a higher layer than TCP/UDP, for example, by repeated transmission. UDP also has the considerable advantage over TCP for signalling that, while TCP loses throughput due to its flow control, UDP has priority in a congested network (although this imbalance is partially addressed by the use of random early detection (RED) [15]).

4. Interconnection with the telephone network

If Internet telephony users can make calls to, or receive calls from, users of ordinary telephones connected to the PSTN, the potential number of calls carried at least part way across the Internet is greatly increased, compared with the situation where both ends need a suitably equipped PC.

It is entirely feasible to implement a gateway between the Internet and the PSTN (although it is less easy to build a gateway that will reliably support the same number of users as even the smallest local telephone exchange). The simplest single-line gateway, literally a PC with a sound card, a modem, and two telephone lines, has been available as freeware since the end of 1995. Subsequently, commercial implementations have been demonstrated or announced by companies with various backgrounds, including both switch manufacturers and speech technology companies.

A gateway requires multiple instances of the IP/resilience/codec stack found in a simple I-Phone, connected to multiple PCM channels through appropriate protocol and audio conversion hardware and software (see Fig 3).

It also requires a suitable IP call-control protocol, with support for multiple simultaneous calls to its IP address (often achieved by transferring to a call-specific address

once initial contact is made) as well as a suitable signalling software stack for establishing the PSTN calls, and a means of ensuring that calls into the PSTN initiated from the Internet can be associated with a customer and charged.

Outgoing calls from the Internet to the PSTN are a relatively simple matter compared with calls in the opposite direction, because it is relatively simple for a screen-based I-Phone application to capture a phone number and (if not built in) the IP address of the gateway. It is entirely credible that, in the absence of regulatory constraints, such an application could even look up a directory service to translate a name and address into the PSTN phone number and gateway IP address for a particular called party.

The reverse direction from PSTN to Internet is difficult because it is impractical to require the PSTN user to enter first the telephone number of a gateway and then the numeric IP address of the called party (who is more likely than not to be off line anyway). Neither the telephone keypad nor voice recognition are ideally suited to entering the more memorable domain name service or e-mail address forms either.

It is therefore likely that PSTN calls routed into the Internet will be diverted from a PSTN number by some form of network intelligence, whether a 'personal assistant' diverting on busy from a user's home phone to a mobile, and I-Phone, and finally if all else fails a messaging service, or just a simple user-controlled diversion.

There are two main scenarios for Internet telephony connection to the PSTN. These are local-end breakout and far-end breakout.

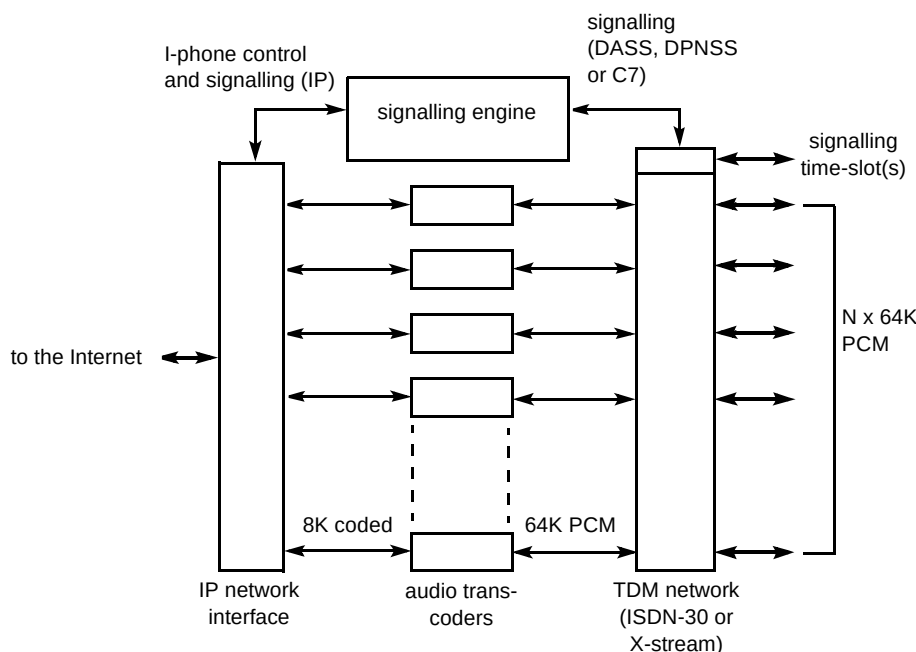


Fig 3 Gateway architecture (logical).

4.1 Local end breakout

Local-end breakout involves the provision of a gateway at an Internet service provider's point of presence (see Fig 4). Telephony traffic is carried with data traffic over a single line connection to the ISP's premises, and is then separated for completion as a normal PSTN call from the gateway to its destination.

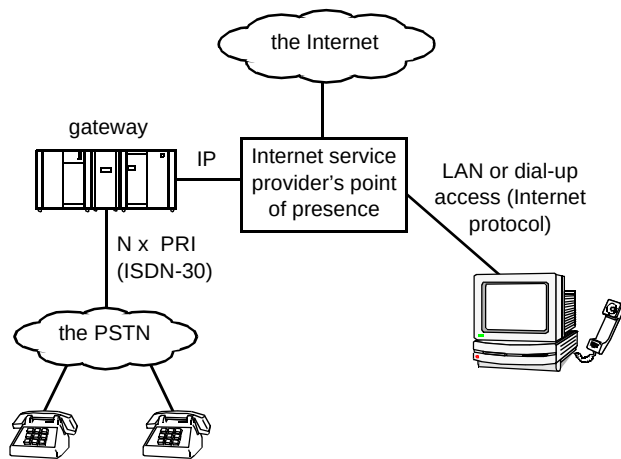


Fig 4 Local-end breakout.

Local-end breakout is essentially a method of sharing a single telephone line between Internet (data) traffic and telephony traffic. Example applications are talking to a help desk without having to suspend an Internet access call, or discussing an on-line (WWW) catalogue with a human sales agent on an 0800 number while still being able to browse the catalogue.

The benefits of this solution to the user are primarily the avoidance of a second PSTN line for outgoing telephone traffic while they are connected to their ISP (or the expense of mobile calls while connected), and the ability to dynamically share their limited access-network bandwidth between voice and data on a second-by-second basis.

Because there is little or no opportunity for congestion between the user's PC and the gateway to cause packet loss, it is possible to achieve near-PSTN quality (with some additional delay) in this scenario. In the longer term, by intelligent routing (see Fig 5), it may be possible to deliver incoming calls to connected Internet users.

A further probable application for this technology is to provide the ability for a World Wide Web user to click directly on a telephone number and be connected, without the loss of their Internet session. This is particularly important for opening up the market for significant telephony call stimulation via WWW advertising and commerce. A typical example would be triggering a call by selecting a WWW hypertext link which would be delivered via the Internet to another computer, or to the PSTN. Note

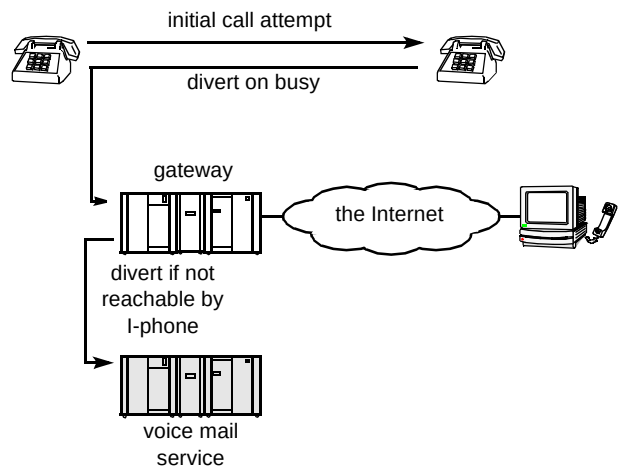


Fig 5 Successive call diversion.

that both mechanisms could be charged equally for equal capability, quality and guarantee.

4.2 Far-end breakout

In the case of far-end breakout (see Fig 6), the majority of the distance between the two parties is over the Internet leg of the call. The motivation for far-end breakout is primarily to avoid conventional long-distance or international call charges by exploiting the flat-rate, subscription-based model of the Internet or a corporate data network based on leased lines (ultimately, the Internet is based on leased cable and satellite connections, so there is little real difference in hardware — just in the commercial model).

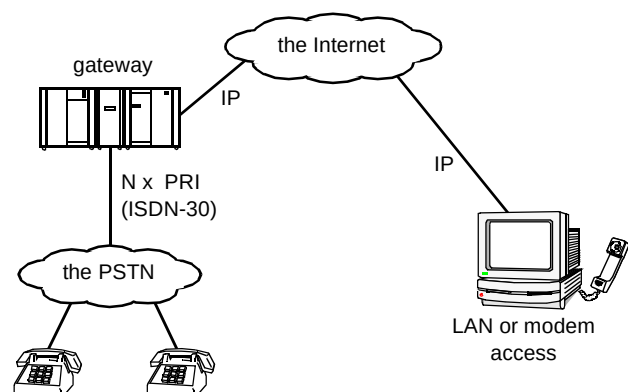


Fig 6 Far-end breakout.

Far-end breakout performance for a commercial public service is limited by the path between caller and gateway over the public Internet. It is therefore likely to be comparable with PC-to-PC I-Phone connection between the same locations.

4.3 Intranets

A special case is where the connection from user to gateway is by a local LAN or WAN. Within a private or managed data network the potential exists to reserve some bandwidth for telephony traffic, or to use data bandwidth for telephony at off-peak times to save on leased circuit-switched voice capacity. Since quality and delay equivalent to a GSM digital mobile phone can readily be achieved on a LAN-based network with I-Phones at their present state of development (at least with a modest number of users), the potential to eliminate building wiring and PBXs for internal calls is obvious and will probably lead to the first business application of packet audio technology, with a gateway for calls outside the building or campus covered by the LAN. Since such a gateway only has to handle external calls, it will be much smaller and cheaper than a PBX for the same user community (which would have to route all calls including internal ones).

It must be emphasised that it is not yet economically worthwhile (and will not be viable for some time) to replace existing separate PBX and LAN installations in this way. It is, however, likely that the economies of an integrated solution will begin to be exploited at new sites within the next few years, probably where extremely tight computer/telephony integration is needed for other, application-specific reasons.

A small extension of this is the concept of a LAN-connected desktop telephone, which has now been proposed by a number of manufacturers. Such a telephone would connect to the corporate LAN, or intranet, rather than dedicated telephone wiring, and remove the need for a PC running I-Phone software to make telephone calls. Calls within the intranet would require little more than an addressing or numbering scheme, but external calls would require a gateway from the intranet to the PSTN.

5. Economics of Internet telephony

5.1 The cost of Internet calls

Internet access offers the user a subscription-based distance-independent 'all you can eat' tariff structure. Most residential users have dial-up access meaning that they also pay a small, local 'per-minute' call charge to connect to their Internet service provider.

Most 'calls' on the UK Internet are international calls for the foreseeable future since the majority of information resources on the Internet reside in the US. In general, a user browsing the WWW pays no attention to the location of the information they retrieve. They are usually more interested

in content than location. Because of this a distance-independent tariff works well.

In the telecommunications arena most calls are local or medium-distance national calls. In general, call volumes are inversely proportional to the distance between the calling and called parties. Because of this it seems sensible to base telephony tariffs around distance and because of the relative rarity of international calls users are charged a premium for making them. International accounting rates provide a means by which the operator originating a call pays for its correspondent in the destination country to deliver it and are reflected in the charges for IDD calls.

Both charging regimes are tailored to the type of service being offered. Internet tariffs are tailored to a network supporting simple non-real-time information retrieval services and IDD tariffs are tailored to a network supporting distance-sensitive real-time services. The addition of real-time services to the Internet offers great opportunities to users to 'freeload' off the Internet. The convergence of real-time and non-real-time services on to the same network challenges the assumptions which were first used to fix Internet tariffs and may ultimately lead to a change in tariffing philosophy.

When packet voice was introduced on to the Internet this changed the perceived value of distance to Internet users. With Internet telephony it is suddenly very significant that a user can communicate with another user anywhere in the world for the same flat rate. However, when the actual costs of carrying calls on the PSTN and on the Internet are compared it is clear that in both cases total cost increases with distance.

The key point is that the costs of carrying an international call over the Internet differ very little from the costs of carrying an IDD call if a similar bandwidth is used.

Table 1 shows that, ignoring overheads, both services use similar amounts of switching and transmission equipment. Both systems can take advantage of low-rate encoding schemes and silence suppression to save on bandwidth, thereby keeping distance-related costs to a minimum and both systems probably incur similar switching costs. The use of more severe coding schemes for packet voice over the Internet can increase more transmission savings at the expense of voice quality. A simple back-of-the-envelope calculation shows that switch/router costs are not particularly significant in the cost equation.

Table 1 Comparison of Internet and IDD telephony services.

	Internet telephony	IDD telephony
Switch costs	10—15 routers + 2 LEs	2 LEs, 2—4 MSUs, 2 ISCs
Transmission costs	2 × 64 k local line 2 × 14.4 k national 2.4 — 8 k transatlantic	2 × 64 k local 2 × 64 k national ~16 k transatlantic
Overheads (support, sale, etc)	One off overhead associated with sales and marketing of software.	Support, management, sales and marketing ~40% of total direct costs.
Billing	Billing costs only incurred by originating telco.	Billing ~5—10% of total direct cost.

A typical switch will see 200 000 BHCA (busy hour call attempts). Assuming 20% of calls in the busy hour this gives an annual calling volume of 365 MBHCA. If the annual costs of such a switch are taken to be £1M then the cost per call per switch is about 0.25p giving a total end-to-end switching cost of about 0.7p/min. Similarly router costs can be gauged from the volume of data produced by an Internet telephony call compared to the annual throughput of say a £10K router capable of supporting data rates in the region of 10 Mbit/s.

A three-minute Internet telephony call would probably occupy about 2 minutes of router time and take up about one thousandth of total throughput. So one call would make up about 5×10^{-8} of annual throughput giving a per router cost for a three-minute call of 0.05p. This gives a total routing cost of about 0.25p/min per call. Estimates of transmission cost, based purely on rentals suggest a per call cost of 0.3p/min per kbit/s of bandwidth occupied, showing that transmission is clearly a significant cost factor. Current Internet telephony is based around 8 kbit/s coding but schemes utilising 2.4 kbit/s (with surprisingly good call quality) have been developed. Such a low rate offers international telephony for a cost of about 1.2 p/min. With 8 kbit/s coding the cost is closer to 3 p/min.

Ultimately it is only the current structure of Internet tariffs, and the lack of support systems and zero billing overheads that allow users to make cheap international calls and not the underlying cost base of the Internet. It is also important to take into account that the user pays an annual fee after which they may (currently) access any service they like. Therefore only users who make regular international calls will see any true cost saving while low users will simply be subsidising those users who do make a saving.

5.2 Hardware costs and costs to end users

Internet telephony is not justifiable except as an incremental cost on top of a computer, and as an Internet connection already purchased and cost-justified against other requirements (although on an intranet, the cost of

upgrading desktop PCs to multimedia specification can be set against a separate PBX and telephony cabling).

Internet telephony is possible using half-duplex soundcards now supplied as standard in most PCs, but this allows only walkie-talkie style conversation which may be considered sufficient by some users for use over the public Internet where potentially long delays would make normal conversation difficult anyway. Full-duplex cards which allow normal conversation currently cost somewhere in the region of £100, but as Internet telephony becomes increasingly popular it is likely that prices will fall, and full-duplex cards will start to be fitted as standard in PCs. The user will also require a microphone and speakers or headset, which are now included with most multimedia PCs, but if bought separately would cost in the region of £20 for speakers/headset and £5 for a microphone, depending on quality.

Some I-Phone software is provided over the Internet free of charge, but most companies charge for the software. The most popular commercial I-Phone package has a suggested list price of \$49.95 for software and licence. The client software for at least one of the Internet-to-PSTN gateway services is offered free of charge, but in this case the user is charged for usage on a per-minute basis.

No-one could justify the cost of a new PC, an ISP account, and local telephone charges accessing the Internet by international call-charge savings, unless they were involved in international calls for several hours daily. The additional cost of a full-duplex soundcard and possibly some software and a microphone, is not large as an increment to a multimedia PC already in service or being purchased for other reasons, and could easily be justifiable for a user who makes a reasonable number of international calls.

5.3 Economics of breakout into the switched network

The costs associated with setting up a PSTN breakout service are almost entirely dependent upon the location and number of Internet/PSTN gateway servers used. Ignoring billing and overhead costs the three main expense areas in setting up such a service are:

- the server IP connection to the 'public Internet',
- the IP/PSTN gateway server,
- interconnect and handover charges to the local PTT operator.

The three cases examined here are near-end server, far-end server and multiple servers.

Near-end breakout

This solution is best used where the service is providing a method of sharing a single telephone line between Internet (data) traffic and telephony traffic. Example applications include talking to a help desk without having to suspend an Internet access call, or discussing an on-line (WWW) catalogue with a human sales agent on an 0800 number while still being able to browse the catalogue. For international calls the hand-over charge levied by the local telecommunications operator will be directly related to the destination and so a near IDD tariff would have to be charged. As such this is not a good way of offering cheap international telephony.

Far-end breakout

This solution provides a good way of offering a restricted low-tariff low-cost international telephony service. The server is located in the destination country of interest or as near as possible to the destination. Since the origin-to-server leg of the call is carried on the public Internet all calls will incur identical costs and tariffs. The cost of connecting the server to the Internet and of providing the server are the same as for near-end breakout.

However, because the server is now closer to the eventual destination the handover charge levied is likely to be as little as 1 or 2 p/min. The main limitation is that really low-tariff calls can only be offered to the customer for PSTN destinations near to the IP/PSTN gateway server.

As long as the traffic volumes to the destination country are high enough, this service would be economical. If the destination country also has a relatively low handover tariff (e.g. the US), then this service can be used to provide a relatively low ISR-type tariff to callers from outside that country to a larger range of international destinations.

Multiple servers

Using multiple servers would allow a provider to offer far-end breakout services to a number of destinations. In order to be economic the number of servers must be carefully matched to the traffic levels generated to their local catchment areas. Extra overheads in terms of routing to the right server are incurred and utilisation of servers may be lower in some areas. In fact, the overall service will be most economic, if each server could be economic as a stand-alone unit. This method allows potential for more revenue, but may in turn have lower margins due to increased overheads.

5.4 *Traditional network versus Internet telephony*

154 The use of packet voice technology does not significantly reduce the cost of carrying a single

international telephone call. Lower rate encoding techniques and the ability to statistically multiplex packets leads to bandwidth savings but can be used in both packet-switched and circuit-switched environments. The lack of need for a complex billing system and any support systems and the lack of any international settlement agreements make for the real difference in tariffs seen between Internet telephony and traditional IDD calls.

In the short term at least, however, the relative inconvenience of international Internet telephony compared with IDD means that it is more of a threat to other cut-price services (such as dial-back) than to conventional IDD.

In the next five to ten years the trend is likely to be towards much lower cost international telephony. Bandwidth prices are falling and coding techniques are improving, leading to reduced data rates and therefore smaller distance-related costs.

Competition in general is likely to lead to lower tariffs and more emphasis on cost saving but this is unlikely to remove the demand for reliable service, customer service support, itemised billing and guaranteed levels of service and call quality.

The ultimate impact and success of Internet telephony depends upon the quality of service and user expectations of the service. In the short term lack of network services, the need to be logged on to a PC before making a call, the need to own a multimedia PC to originate calls, the inability to receive incoming calls from PSTN phones, the lack of customer fault support and the lack of service guarantees are likely to hold back Internet telephony growth.

Ultimately traffic is going to migrate from the traditional PSTN network on to a number of national and international broadband multi-service (packet) networks. The key to long-term success in the communications market is to migrate the traffic in the most efficient way and provide the most appropriate communications solutions to the end user.

6. **Technical trends**

In the short term, Internet telephony growth is likely to be based on optimisation of the current technology described in this paper. The growth limits for current terminal equipment and network implementations will depend on network capacity growth (lower bound) and the number of potential PC terminals (upper bound).

There are four significant factors that are likely to determine the eventual scale of Internet telephony deployment.

- Network growth

Network providers like BT and MCI have announced planned increases of between three and five times in their Internet backbone network capacities over 1996-97. Much of this is driven by growth in the number of users, and demand for traditional services such as the World Wide Web and e-mail, so it would be unsafe to assume that all (or even most) of the extra is available for telephony.

- Improved compression

The Internet telephony, mobile radio, and military markets have driven the development audio codecs offering acceptable quality at 2.4 kbit/s ($\frac{1}{3}$ the bandwidth required by established I-Phone products) while running on a Pentium or equivalent processor.

- New services

It is likely that telephony will have to share the Internet with an increasing number of real-time and non-real-time services. Electronic commerce, multicasting, on-line 'radio' and 'television' stations, and ever-more-complex World Wide Web sites will all be growing into the enlarged backbone network along with telephony.

For the foreseeable future it is likely that packet audio (whether as Internet telephony or in more specialised forms) will be associated with other multimedia services such as multi-player games rather than straight telephony.

- Network enhancements

The Internet Engineering Task Force (IETF) is developing new protocols and technologies to support real-time services, bandwidth reservation, and potentially per-session or per-call charging. These are discussed elsewhere [1, 11].

7. Conclusions

When discussing the case for Internet telephony, it is important to consider all the factors driving its development. Much has been made of the ability to bypass international carriers and the associated accounting rate payments as described above. However, when analysing the fundamental costs it can be seen that Internet telephony of equal service specification does not have any significant cost advantages over traditional telephony infrastructure. However, this is not the complete picture and all the factors and trade-offs driving the use of Internet telephony need to be assessed.

Having considered cost, the second issue is both quality, and the type of quality guarantees. It is now possible to provide Internet telephony of very similar quality to that of

the PSTN. This is easily achieved away from the traffic peak (unguaranteed) or through the use of priority buffering and signalling (guaranteed). During peak periods, the unguaranteed user may be happy to try again later, particularly if it is for unconventional audio applications such as chat and computer gaming. The business user, whose reputation or timeliness could be affected by unguaranteed performance may be prepared to pay the cost of the guarantee. The trade-off here is that a shared network, with no telephony operational support systems (OSS) and other overheads, should be significantly cheaper to use for telephony than a telephony-specific network with OSS, but that this network will exhibit neither guaranteed availability, quality nor auditable usage. If that infrastructure can also support the guaranteed traffic then the user is presented with all the options from which they can select the appropriate cost/quality combination for the task in hand.

On a different tack, it is also possible to provide either higher or lower bandwidth audio over this same infrastructure, again depending on the cost versus quality trade-off. While the PSTN is limited to 64K switching (although expensive network-based compression and audio processing can be used to extend this range), the Internet is already able to offer the user a range of bandwidths from 4 kbit/s up to stereo audio, and this range of capabilities will be expanded at incremental cost, as various types of end-to-end bearer qualities and audio-processing components become available as part of the Internet integrated services architecture.

The range of audio capabilities is an example of the flexibility of the infrastructure and the CPE, which is the third concept to consider. The computer can provide a number of features at incremental cost leading to the integration of a number of processes which have traditionally required separate CPE. The standard computer is already coming equipped with modems for Internet access. These will also support the sending of faxes through the addition of fax software, and faxes can then be received, stored, processed and printed in a very flexible way. The addition of audio drivers to computers is essential for a number of other application processes such as audio-enabled WWW, speech recognition and playing CDs. The receipt of telephony via traditional telephony interfaces to the computer is already evident which enables voice and video messages to be recorded, processed, and linked to data records on the computer network. There is therefore a strong trend towards computer-based telephony, but data and audio are still treated separately using different interfaces. Similarly, the Internet provides a very rich mixture of application and transport capabilities which can be mixed and matched to provide useful services. Most

importantly though it provides a means to cope with a range of multimedia traffic with wide-ranging transport requirements which enables integration at the desktop.

The combination of software-based CPE and network functionalities leads to a highly flexibly and fast-moving user capability, against which the traditional 64K switched network and the simple telephone must be compared. It is already becoming apparent that the computer and multimedia networking could usurp many of the roles of the PBX and the intelligent network platforms in the future due to this flexibility and integration capability.

The final reason for the inevitable growth of Internet telephony is that of convenience. The user is working on their computer terminal, possibly surfing the Internet, and does not want to be prevented from either sending or receiving telephony calls. A telco could take a position that this should be solved by the user buying a second line but clearly this is a significant commercial barrier and does not provide application integration unless both lines appear on the computer terminal. The user would clearly like to have the option to suspend surfing, and send a call without having to log out of the Internet and swap over connections. This is particularly important for opening up the market for significant telephony call stimulation via WWW advertising and commerce. Network-based answering machines can reduce the burden of missing incoming calls but it would clearly be beneficial if the user had the option to receive the call directly via a message to the computer screen. This argument also suggests that the user will not wish to switch on their computer before receiving and making telephone calls. It is therefore apparent that computer/telephony interfaces need to evolve to enable the immediate reception and making of calls while also enabling the user to run parallel, competing but prioritised sessions through the single PSTN network interface. In addition, the user would also like to be able to buy that second line or upgrade to another network technology, and have all the bandwidth available at the desktop for the range of tasks in hand. The challenge ahead is determining which one, or combination of, the future infrastructure and protocol options can support the cost versus quality versus flexibility versus convenience trade-offs most effectively.

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